$A^TAx = A^Tb$

$x = (A^TA)^{-1}A^T b$

國立陽明交通大學 115 學年度碩士班考試入學招生試題

projection matrix?

$A(A^TA)^{-1}A^T$

科目：線性代數與離散數學(8102)　　考試日期：115 年 2 月 4 日　第 2 節

系所班別：資訊聯招　　

【不可使用計算機】＊作答前請先核對試題、答案卷(試卷)與准考證之所組別與考科是否相符！！

1. True or False: This part consists of true-or-false questions. In each case, please answer true if the statement is always true and false otherwise. And give a counterexample for any false conjecture.

1.1. **(5 points)** A *nxn* matrix **A** is said to be nonsingular,

F (1) if **A** is non-invertible. counter-ex. :

T (2) if there exists a matrix **B** such that **AB**=**BA**=**I**.

T (3) if the determinates of **A** in nonzero.

F (4) if **Ax**=**0** where x is nonzero vector. A is singular

1.2. **(5 points)** There are many types of elementary matrices. Which of the following definitions are elementary matrices?

T (1) A *nxn* matrix is obtained by interchanging two rows of **I**.

T (2) A *nxn* matrix is obtained by multiplying a row of **I** by a nonzero constant.

T (3) A *nxn* matrix is obtained from **I** by adding a multiple of on row to another row.

F (4) A *nxn* matrix is form from the **I** by reordering its column.

1.3. **(5 points)** If there are two *nxn* matrices **A**, **B** and one scalar *r*,

F (1) det(**A**) = det(**B**) implies **A** = **B**.

F (2) det(*r***A**+**B**) = *r* det(*r***A**)+det(**B**).

T (3) det((**AB**)^T) = det(**A**)det(**B**).

T (4) det(**A**) = det(**B**) where **A** and **B** are row equivalent matrices.

1.4. **(5 points)** The $\mathbf{a}_1, \mathbf{a}_2, \ldots, \mathbf{a}_n$ be n column vectors in R^n and let $\mathbf{A} = [\mathbf{a}_1\ \mathbf{a}_2 \ldots \mathbf{a}_n]$.

F (1) The vectors $\mathbf{a}_1, \mathbf{a}_2, \ldots, \mathbf{a}_n$ will be linearly independent if and only if **A** is ~~singular.~~ non-singular

T (2) The **Ax** = **0** if and only if **x** is trivial $[0\ 0 \ldots 0]^T$.

T (3) The rank of **A** plus the nullity of A equals *n*. R(A)+N(A)

T (4) The rank of **A** equals the rank of $\mathbf{A}^T$.

$\begin{bmatrix} a & b \\ c & -a \end{bmatrix}$

2. **(5 points)** By trial and error, find the general condition of 2×2 matrices **A**, such that $\mathbf{A}^2 = -\mathbf{I}$ with real entries.

$a^2+bc=-1$

$A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}\begin{bmatrix} a & b \\ c & d \end{bmatrix} = \begin{bmatrix} a^2+bc & ab+bd \\ ac+cd & bc+d^2 \end{bmatrix}$

$\begin{cases} a^2+bc=-1 \\ b(a+d)=0 \\ c(a+d)=0 \\ bc+d^2=-1 \end{cases}$　① $b=0$, ✱　② $c=0$, ✱　③ $a+d=0$, $b\neq0$, $c\neq0$

$\boxed{a^2=d^2}$　$a=\pm d$　⇓ $a=-d$

3. **(5 points)** Find the determinant of the matrix $A = \begin{bmatrix} 1 & 0 & 2 & 0 & 0 & 4 \\ 30 & 1 & 5 & 0 & 0 & 9 \\ 3 & 5 & 3 & 8 & 0 & 4 \\ 2 & 0 & 8 & 0 & 0 & 7 \\ 7 & 0 & 4 & 0 & 6 & 0 \\ 0 & 0 & 1 & 0 & 2 & 0 \end{bmatrix}$

$\det\begin{bmatrix} A & B \\ C & D \end{bmatrix} = \det(A)\det(D) - \det(B)\det(C)$

$= 272 \cdot 0 - 0 \cdot 0 = 0$

X 4. **(8 points)** Find a Singular Value Decomposition of the matrix $A = \begin{bmatrix} 2 & -2 \\ 2 & 2 \\ 1 & -1 \end{bmatrix}$ and verify your answer.

$x^Tx=1$

5. **(12 points)** Let A be an mxn matrix and x be an nx1 vector. (a) (4 points) Given $\|x\| = 1$, what is the least square solution for $Ax = 0$? (b) (8 points) Please prove your answer. (Please leave it blank if you don't know the correct answer, or you will get **at most minus 5 points** (until questions 3, 4, and 5 are 0 points) for the wrong answer. 不會寫請留白，答錯最多倒扣 5 分，扣至 3, 4, 5 題 0 分為止。)

$A^TAx = A^Tb$　　$(x^T)^{-1} = (A^TA)^{-1}A^Tb$

$x = (A^TA)^{-1}A^Tb$　　$(x^{-1})^T = (A^TA)^{-1}A^Tb$

$1 = x^Tx = x^T(A^TA)^{-1}A^Tb$　　$x^{-1} = (A^Tb)^T[(A^TA)^{-1}]^T = (b^TA)(A^TA)^{-1}$

294
− 22
272

$\det(A) = \begin{vmatrix} 1 & 0 & 2 \\ 30 & 1 & 5 \\ 3 & 5 & 3 \end{vmatrix} = \begin{vmatrix} 1 & 5 \\ 5 & 3 \end{vmatrix} + 2\begin{vmatrix} 30 & 1 \\ 3 & 5 \end{vmatrix} = -22 + 2(150-3) = -22 + 2 \cdot 147 = 272$

$\det(B) = 0$, $\det(C) = 0$, $\det(D) = 0$

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6. There are five independent subproblems in this question.

$\frac{1-(\frac{1}{2})^k}{1-\frac{1}{2}} = 2(1-(\frac{1}{2})^k)$

(6.1) (5 points) Is the following statement a tautology, a contradiction, or neither? You need to provide a rigirous explanation or proof.

negate is ω

$\frac{(\frac{1}{2})^k-1}{\frac{1}{2}-1} =$

$$\nexists a.\forall b(b \in a \leftrightarrow b \notin b)$$

a	b	a↔b
T	T	T
T	F	F
F	T	F
F	F	T

$\neg\exists a\, \forall b$

$\neg \forall a \exists b\ (b \notin a \leftrightarrow b \in b)$

(6.2) (5 points) What is x in the following equation? (Assume the left-hand-side term of the equation converges.)

$$\Sigma_{r=2}^{\infty}(2^{x-r}) = \sqrt{1+3\times 2^{x-2}}$$

$\frac{2^x}{2^r}$　$2^x/4,\ 2^x/8,\ 2^x/16 \cdots$　$2^x/2^\infty \approx 0$

$2^x(\frac{1}{4}+\frac{1}{8}+\frac{1}{16}+\cdots)$

$= 2^{x-2}(1+\frac{1}{2}+\frac{1}{4}+\cdots) = 2^{x-2}$

You need to show how to obtain your answer.

(6.3) (5 points) What is the integer part of $\sqrt{5}+\sqrt{6}+\sqrt{7}+\sqrt{8}+\sqrt{9}+\sqrt{10}+\sqrt{11}+\sqrt{12}+\sqrt{13}$? You need to show how to obtain your answer.

$(2^2, 2^3) = \sqrt{5}, \sqrt{6}, \sqrt{7}$,　$2^3 = 8$,　$(2^3, 2^4) \Rightarrow \sqrt{10}, \sqrt{11}, \sqrt{12}, \sqrt{13}$

(6.4) (5 points) Let $T =_{def} 11\ldots11$ (there are 243 1's). Is T divisible by 243? You need to explain your answer.

$\underbrace{11\cdots1}_{243}$

$243_{10} = 11110011_2$

(6.5) (5 points) Given a strictly increasing function $f : N \to N$ so that $f(f(x)) = 2x+1$, what is $f(6)$? Note that N denotes the set of natural numbers.

2 | 243　1
2 | 121　1
3 | 60　0
2 | 30　0
2 | 15　1
2 | 7　1
2 | 3　1
　1

$f \circ f$

3, 4, 5, …

$f(6) = \sqrt{2}\cdot 6 + \sqrt{2} - 1$
$= 7\sqrt{2} - 1$

$f(6)$　$f(6)$

2x+1

$f(x) = x$

$\frac{2^n - 1}{2-1}$

$f(x) = 6$

$f(x) = ax + b$

$f(f(x)) = a f(x) + b$
$= a(ax+b) + b$
$= a^2x + (ab+b)$

$a = \sqrt{2}$

$(1+\sqrt{2})b = 1,\ b = \frac{1}{1+\sqrt{2}}$

$\frac{\sqrt{2}-1}{(\sqrt{2}+1)(\sqrt{2}-1)} = \sqrt{2}-1$

1+1+1

$2\sqrt{2} = 2\cdot(1.414)$
$= 2.828$

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7. For $n \geq 0$, let $\{x_n\}, \{y_n\}, \{z_n\}$ be sequences defined by the initial conditions $x_0 = 1$, $y_0 = 0$, and $z_0 = 0$. For $n \geq 1$, they satisfy $x_n = x_{n-1} + y_{n-1} + z_{n-1}$, $y_n = x_{n-1} + z_{n-1}$, and $z_n = y_{n-1}$. You must show all necessary steps. **Answers without justification will receive no points.**

(a) **(3 points)** Let $u_n = x_n + y_n + z_n$. Use mathematical induction to prove that $u_n = 2^n$ for all $n \geq 0$.

(b) **(3 points)** Using the result from part (a), derive a recurrence relation for y_n in terms of y_{n-1} and show that $y_n = 2^{n-1} - y_{n-1}$ for all $n \geq 1$.

(c) **(4 points)** Solve the recurrence in part (b) to obtain a closed-form expression for y_n, and verify that your formula satisfies the initial condition.

8. Define relations R and S on $\mathbb{N} = \{0,1,2,\ldots\}$ by (1) xRy if and only if $x \equiv y \bmod 7$, and (2) xSy if and only if $x \equiv y \bmod 13$. For each of the following relations on $\mathbb{N}$, determine whether it is an equivalence relation. If it is, give a proof. If it is not, explain which property of an equivalence relation fails. In each case, justify your answer. **Answers without justification will receive no points.**

trans. , anti.-symm , reflexive

(a) **(4 points)** $R \cap S$　ϕ

(b) **(4 points)** $R \cup S$

$\begin{cases} x \equiv y \pmod 7 \\ x \equiv y \pmod{13} \end{cases}$

9. **(7 points)** A fully connected communication network on n nodes is modeled by the complete graph K_n, where $n \geq 5$. Let $S \subseteq V(K_n) \cup E(K_n)$ be a set of failed nodes and links with $|S| \leq n-3$, such that no failed link is incident to a failed node. Let $\boxed{K_n - S}$ denote the network obtained after removing all failed nodes (and their incident links) and all failed links. Prove by induction on n that the remaining network $K_n - S$ contains a Hamiltonian cycle. **You may not assume any characterization theorem for Hamiltonian graphs.**

(a)

[Base] $u_0 = x_0 + y_0 + z_0 = 1+0+0 = 1 = 2^0$

assume k, $u_k = x_k + y_k + z_k = 2^k$

when $k+1$, $u_{k+1} = x_{k+1} + y_{k+1} + z_{k+1}$

$= (x_k + y_k + z_k) + (x_k + z_k) + y_k$

$= 2(x_k + y_k + z_k)$

$= 2 \cdot 2^k = 2^{k+1}$

(b) $\begin{cases} x_n = x_{n-1} + y_{n-1} + z_{n-1} \\ y_n = x_{n-1} + z_{n-1} \end{cases}$

$x_n - y_n = y_{n-1}$, $x_n = y_n + y_{n-1}$

$2^n = u_n = x_n + y_n + z_n$

$= (y_n + y_{n-1}) + y_n + (y_{n-1}) = 2(y_n + y_{n-1})$

$2^{n-1} = y_n + y_{n-1}$

$y_n = 2^{n-1} - y_{n-1}$

(c)

$y_1 = 1 + 0 = 1$

$y_n = -y_{n-1}$

$\cancel{x+1=0}$, $x = (-1)^n$, $y_n^{(H)} = (-1)^n$ 0

$y_0 = -c_1 = 1$, $c_1 = (-1)$

$x = (-1)^{n+1}$

$y_n = -y_{n-1}$

x